

Homework 5

Joseph Shin

2025-09-19

```
crime_data = read.table('uscrime.txt', header = TRUE, sep = '\t')
crime_data
```

##	M	So	Ed	Po1	Po2	LF	M.F	Pop	NW	U1	U2	Wealth	Ineq	Prob
## 1	15.1	1	9.1	5.8	5.6	0.510	95.0	33	30.1	0.108	4.1	3940	26.1	0.084602
## 2	14.3	0	11.3	10.3	9.5	0.583	101.2	13	10.2	0.096	3.6	5570	19.4	0.029599
## 3	14.2	1	8.9	4.5	4.4	0.533	96.9	18	21.9	0.094	3.3	3180	25.0	0.083401
## 4	13.6	0	12.1	14.9	14.1	0.577	99.4	157	8.0	0.102	3.9	6730	16.7	0.015801
## 5	14.1	0	12.1	10.9	10.1	0.591	98.5	18	3.0	0.091	2.0	5780	17.4	0.041399
## 6	12.1	0	11.0	11.8	11.5	0.547	96.4	25	4.4	0.084	2.9	6890	12.6	0.034201
## 7	12.7	1	11.1	8.2	7.9	0.519	98.2	4	13.9	0.097	3.8	6200	16.8	0.042100
## 8	13.1	1	10.9	11.5	10.9	0.542	96.9	50	17.9	0.079	3.5	4720	20.6	0.040099
## 9	15.7	1	9.0	6.5	6.2	0.553	95.5	39	28.6	0.081	2.8	4210	23.9	0.071697
## 10	14.0	0	11.8	7.1	6.8	0.632	102.9	7	1.5	0.100	2.4	5260	17.4	0.044498
## 11	12.4	0	10.5	12.1	11.6	0.580	96.6	101	10.6	0.077	3.5	6570	17.0	0.016201
## 12	13.4	0	10.8	7.5	7.1	0.595	97.2	47	5.9	0.083	3.1	5800	17.2	0.031201
## 13	12.8	0	11.3	6.7	6.0	0.624	97.2	28	1.0	0.077	2.5	5070	20.6	0.045302
## 14	13.5	0	11.7	6.2	6.1	0.595	98.6	22	4.6	0.077	2.7	5290	19.0	0.053200
## 15	15.2	1	8.7	5.7	5.3	0.530	98.6	30	7.2	0.092	4.3	4050	26.4	0.069100
## 16	14.2	1	8.8	8.1	7.7	0.497	95.6	33	32.1	0.116	4.7	4270	24.7	0.052099
## 17	14.3	0	11.0	6.6	6.3	0.537	97.7	10	0.6	0.114	3.5	4870	16.6	0.076299
## 18	13.5	1	10.4	12.3	11.5	0.537	97.8	31	17.0	0.089	3.4	6310	16.5	0.119804
## 19	13.0	0	11.6	12.8	12.8	0.536	93.4	51	2.4	0.078	3.4	6270	13.5	0.019099
## 20	12.5	0	10.8	11.3	10.5	0.567	98.5	78	9.4	0.130	5.8	6260	16.6	0.034801
## 21	12.6	0	10.8	7.4	6.7	0.602	98.4	34	1.2	0.102	3.3	5570	19.5	0.022800
## 22	15.7	1	8.9	4.7	4.4	0.512	96.2	22	42.3	0.097	3.4	2880	27.6	0.089502
## 23	13.2	0	9.6	8.7	8.3	0.564	95.3	43	9.2	0.083	3.2	5130	22.7	0.030700
## 24	13.1	0	11.6	7.8	7.3	0.574	103.8	7	3.6	0.142	4.2	5400	17.6	0.041598
## 25	13.0	0	11.6	6.3	5.7	0.641	98.4	14	2.6	0.070	2.1	4860	19.6	0.069197
## 26	13.1	0	12.1	16.0	14.3	0.631	107.1	3	7.7	0.102	4.1	6740	15.2	0.041698
## 27	13.5	0	10.9	6.9	7.1	0.540	96.5	6	0.4	0.080	2.2	5640	13.9	0.036099
## 28	15.2	0	11.2	8.2	7.6	0.571	101.8	10	7.9	0.103	2.8	5370	21.5	0.038201
## 29	11.9	0	10.7	16.6	15.7	0.521	93.8	168	8.9	0.092	3.6	6370	15.4	0.023400
## 30	16.6	1	8.9	5.8	5.4	0.521	97.3	46	25.4	0.072	2.6	3960	23.7	0.075298
## 31	14.0	0	9.3	5.5	5.4	0.535	104.5	6	2.0	0.135	4.0	4530	20.0	0.041999
## 32	12.5	0	10.9	9.0	8.1	0.586	96.4	97	8.2	0.105	4.3	6170	16.3	0.042698
## 33	14.7	1	10.4	6.3	6.4	0.560	97.2	23	9.5	0.076	2.4	4620	23.3	0.049499
## 34	12.6	0	11.8	9.7	9.7	0.542	99.0	18	2.1	0.102	3.5	5890	16.6	0.040799
## 35	12.3	0	10.2	9.7	8.7	0.526	94.8	113	7.6	0.124	5.0	5720	15.8	0.020700
## 36	15.0	0	10.0	10.9	9.8	0.531	96.4	9	2.4	0.087	3.8	5590	15.3	0.006900
## 37	17.7	1	8.7	5.8	5.6	0.638	97.4	24	34.9	0.076	2.8	3820	25.4	0.045198
## 38	13.3	0	10.4	5.1	4.7	0.599	102.4	7	4.0	0.099	2.7	4250	22.5	0.053998

```

## 39 14.9 1 8.8 6.1 5.4 0.515 95.3 36 16.5 0.086 3.5 3950 25.1 0.047099
## 40 14.5 1 10.4 8.2 7.4 0.560 98.1 96 12.6 0.088 3.1 4880 22.8 0.038801
## 41 14.8 0 12.2 7.2 6.6 0.601 99.8 9 1.9 0.084 2.0 5900 14.4 0.025100
## 42 14.1 0 10.9 5.6 5.4 0.523 96.8 4 0.2 0.107 3.7 4890 17.0 0.088904
## 43 16.2 1 9.9 7.5 7.0 0.522 99.6 40 20.8 0.073 2.7 4960 22.4 0.054902
## 44 13.6 0 12.1 9.5 9.6 0.574 101.2 29 3.6 0.111 3.7 6220 16.2 0.028100
## 45 13.9 1 8.8 4.6 4.1 0.480 96.8 19 4.9 0.135 5.3 4570 24.9 0.056202
## 46 12.6 0 10.4 10.6 9.7 0.599 98.9 40 2.4 0.078 2.5 5930 17.1 0.046598
## 47 13.0 0 12.1 9.0 9.1 0.623 104.9 3 2.2 0.113 4.0 5880 16.0 0.052802
##      Time Crime
## 1 26.2011 791
## 2 25.2999 1635
## 3 24.3006 578
## 4 29.9012 1969
## 5 21.2998 1234
## 6 20.9995 682
## 7 20.6993 963
## 8 24.5988 1555
## 9 29.4001 856
## 10 19.5994 705
## 11 41.6000 1674
## 12 34.2984 849
## 13 36.2993 511
## 14 21.5010 664
## 15 22.7008 798
## 16 26.0991 946
## 17 19.1002 539
## 18 18.1996 929
## 19 24.9008 750
## 20 26.4010 1225
## 21 37.5998 742
## 22 37.0994 439
## 23 25.1989 1216
## 24 17.6000 968
## 25 21.9003 523
## 26 22.1005 1993
## 27 28.4999 342
## 28 25.8006 1216
## 29 36.7009 1043
## 30 28.3011 696
## 31 21.7998 373
## 32 30.9014 754
## 33 25.5005 1072
## 34 21.6997 923
## 35 37.4011 653
## 36 44.0004 1272
## 37 31.6995 831
## 38 16.6999 566
## 39 27.3004 826
## 40 29.3004 1151
## 41 30.0001 880
## 42 12.1996 542
## 43 31.9989 823
## 44 30.0001 1030

```

```
## 45 32.5996 455
## 46 16.6999 508
## 47 16.0997 849
```

Use regression (a useful R function is `lm` or `glm`) to predict the observed crime rate in a city with the following data:

M = 14.0 So = 0 Ed = 10.0 Po1 = 12.0 Po2 = 15.5 LF = 0.640 M.F = 94.0 Pop = 150 NW = 1.1 U1 = 0.120 U2 = 3.6 Wealth = 3200 Ineq = 20.1 Prob = 0.04 Time = 39.0 Show your model (factors used and their coefficients), the software output, and the quality of fit. Note that because there are only 47 data points and 15 predictors, you'll probably notice some overfitting. We'll see ways of dealing with this sort of problem later in the course.

```
crime_data = crime_data[-c(4, 26),]
crime_data
```

##		M	So	Ed	Po1	Po2	LF	M.F	Pop	NW	U1	U2	Wealth	Ineq	Prob
## 1	15.1	1	9.1	5.8	5.6	0.510	95.0	33	30.1	0.108	4.1	3940	26.1	0.084602	
## 2	14.3	0	11.3	10.3	9.5	0.583	101.2	13	10.2	0.096	3.6	5570	19.4	0.029599	
## 3	14.2	1	8.9	4.5	4.4	0.533	96.9	18	21.9	0.094	3.3	3180	25.0	0.083401	
## 5	14.1	0	12.1	10.9	10.1	0.591	98.5	18	3.0	0.091	2.0	5780	17.4	0.041399	
## 6	12.1	0	11.0	11.8	11.5	0.547	96.4	25	4.4	0.084	2.9	6890	12.6	0.034201	
## 7	12.7	1	11.1	8.2	7.9	0.519	98.2	4	13.9	0.097	3.8	6200	16.8	0.042100	
## 8	13.1	1	10.9	11.5	10.9	0.542	96.9	50	17.9	0.079	3.5	4720	20.6	0.040099	
## 9	15.7	1	9.0	6.5	6.2	0.553	95.5	39	28.6	0.081	2.8	4210	23.9	0.071697	
## 10	14.0	0	11.8	7.1	6.8	0.632	102.9	7	1.5	0.100	2.4	5260	17.4	0.044498	
## 11	12.4	0	10.5	12.1	11.6	0.580	96.6	101	10.6	0.077	3.5	6570	17.0	0.016201	
## 12	13.4	0	10.8	7.5	7.1	0.595	97.2	47	5.9	0.083	3.1	5800	17.2	0.031201	
## 13	12.8	0	11.3	6.7	6.0	0.624	97.2	28	1.0	0.077	2.5	5070	20.6	0.045302	
## 14	13.5	0	11.7	6.2	6.1	0.595	98.6	22	4.6	0.077	2.7	5290	19.0	0.053200	
## 15	15.2	1	8.7	5.7	5.3	0.530	98.6	30	7.2	0.092	4.3	4050	26.4	0.069100	
## 16	14.2	1	8.8	8.1	7.7	0.497	95.6	33	32.1	0.116	4.7	4270	24.7	0.052099	
## 17	14.3	0	11.0	6.6	6.3	0.537	97.7	10	0.6	0.114	3.5	4870	16.6	0.076299	
## 18	13.5	1	10.4	12.3	11.5	0.537	97.8	31	17.0	0.089	3.4	6310	16.5	0.119804	
## 19	13.0	0	11.6	12.8	12.8	0.536	93.4	51	2.4	0.078	3.4	6270	13.5	0.019099	
## 20	12.5	0	10.8	11.3	10.5	0.567	98.5	78	9.4	0.130	5.8	6260	16.6	0.034801	
## 21	12.6	0	10.8	7.4	6.7	0.602	98.4	34	1.2	0.102	3.3	5570	19.5	0.022800	
## 22	15.7	1	8.9	4.7	4.4	0.512	96.2	22	42.3	0.097	3.4	2880	27.6	0.089502	
## 23	13.2	0	9.6	8.7	8.3	0.564	95.3	43	9.2	0.083	3.2	5130	22.7	0.030700	
## 24	13.1	0	11.6	7.8	7.3	0.574	103.8	7	3.6	0.142	4.2	5400	17.6	0.041598	
## 25	13.0	0	11.6	6.3	5.7	0.641	98.4	14	2.6	0.070	2.1	4860	19.6	0.069197	
## 27	13.5	0	10.9	6.9	7.1	0.540	96.5	6	0.4	0.080	2.2	5640	13.9	0.036099	
## 28	15.2	0	11.2	8.2	7.6	0.571	101.8	10	7.9	0.103	2.8	5370	21.5	0.038201	
## 29	11.9	0	10.7	16.6	15.7	0.521	93.8	168	8.9	0.092	3.6	6370	15.4	0.023400	
## 30	16.6	1	8.9	5.8	5.4	0.521	97.3	46	25.4	0.072	2.6	3960	23.7	0.075298	
## 31	14.0	0	9.3	5.5	5.4	0.535	104.5	6	2.0	0.135	4.0	4530	20.0	0.041999	
## 32	12.5	0	10.9	9.0	8.1	0.586	96.4	97	8.2	0.105	4.3	6170	16.3	0.042698	
## 33	14.7	1	10.4	6.3	6.4	0.560	97.2	23	9.5	0.076	2.4	4620	23.3	0.049499	
## 34	12.6	0	11.8	9.7	9.7	0.542	99.0	18	2.1	0.102	3.5	5890	16.6	0.040799	
## 35	12.3	0	10.2	9.7	8.7	0.526	94.8	113	7.6	0.124	5.0	5720	15.8	0.020700	
## 36	15.0	0	10.0	10.9	9.8	0.531	96.4	9	2.4	0.087	3.8	5590	15.3	0.006900	
## 37	17.7	1	8.7	5.8	5.6	0.638	97.4	24	34.9	0.076	2.8	3820	25.4	0.045198	
## 38	13.3	0	10.4	5.1	4.7	0.599	102.4	7	4.0	0.099	2.7	4250	22.5	0.053998	
## 39	14.9	1	8.8	6.1	5.4	0.515	95.3	36	16.5	0.086	3.5	3950	25.1	0.047099	

##	40	14.5	1	10.4	8.2	7.4	0.560	98.1	96	12.6	0.088	3.1	4880	22.8	0.038801
##	41	14.8	0	12.2	7.2	6.6	0.601	99.8	9	1.9	0.084	2.0	5900	14.4	0.025100
##	42	14.1	0	10.9	5.6	5.4	0.523	96.8	4	0.2	0.107	3.7	4890	17.0	0.088904
##	43	16.2	1	9.9	7.5	7.0	0.522	99.6	40	20.8	0.073	2.7	4960	22.4	0.054902
##	44	13.6	0	12.1	9.5	9.6	0.574	101.2	29	3.6	0.111	3.7	6220	16.2	0.028100
##	45	13.9	1	8.8	4.6	4.1	0.480	96.8	19	4.9	0.135	5.3	4570	24.9	0.056202
##	46	12.6	0	10.4	10.6	9.7	0.599	98.9	40	2.4	0.078	2.5	5930	17.1	0.046598
##	47	13.0	0	12.1	9.0	9.1	0.623	104.9	3	2.2	0.113	4.0	5880	16.0	0.052802
##				Time	Crime										
##	1	26.2011		791											
##	2	25.2999		1635											
##	3	24.3006		578											
##	5	21.2998		1234											
##	6	20.9995		682											
##	7	20.6993		963											
##	8	24.5988		1555											
##	9	29.4001		856											
##	10	19.5994		705											
##	11	41.6000		1674											
##	12	34.2984		849											
##	13	36.2993		511											
##	14	21.5010		664											
##	15	22.7008		798											
##	16	26.0991		946											
##	17	19.1002		539											
##	18	18.1996		929											
##	19	24.9008		750											
##	20	26.4010		1225											
##	21	37.5998		742											
##	22	37.0994		439											
##	23	25.1989		1216											
##	24	17.6000		968											
##	25	21.9003		523											
##	27	28.4999		342											
##	28	25.8006		1216											
##	29	36.7009		1043											
##	30	28.3011		696											
##	31	21.7998		373											
##	32	30.9014		754											
##	33	25.5005		1072											
##	34	21.6997		923											
##	35	37.4011		653											
##	36	44.0004		1272											
##	37	31.6995		831											
##	38	16.6999		566											
##	39	27.3004		826											
##	40	29.3004		1151											
##	41	30.0001		880											
##	42	12.1996		542											
##	43	31.9989		823											
##	44	30.0001		1030											
##	45	32.5996		455											
##	46	16.6999		508											
##	47	16.0997		849											

From Homework 3, I removed two outliers from this exact dataset. The outliers that I removed were when the crime rate was at 1993 and 1969. Therefore, I chose to remove those two observations in this dataset before I perform regression because it may interfere with my analysis.

For AIC, we should use the corrected AIC because it is a small dataset. Look for smaller AIC for better fit.

```
library(stats)
lin_mod = lm(Crime ~ ., data = crime_data)
summary(lin_mod)

##
## Call:
## lm(formula = Crime ~ ., data = crime_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -381.84  -98.79   -6.45   106.81   536.88
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.265e+03  1.880e+03  -2.801  0.00897 **
## M              7.363e+01  4.442e+01   1.658  0.10819
## So              7.028e+00  1.509e+02   0.047  0.96318
## Ed              1.721e+02  6.454e+01   2.666  0.01241 *
## Po1             1.964e+02  1.183e+02   1.660  0.10763
## Po2            -1.152e+02  1.262e+02  -0.913  0.36900
## LF             -5.473e+02  1.492e+03  -0.367  0.71646
## M.F             1.449e+01  2.205e+01   0.657  0.51645
## Pop            -1.468e+00  1.560e+00  -0.941  0.35440
## NW              4.452e+00  6.584e+00   0.676  0.50430
## U1             -5.795e+03  4.368e+03  -1.327  0.19499
## U2              1.634e+02  8.574e+01   1.905  0.06672 .
## Wealth          8.985e-02  1.061e-01   0.847  0.40412
## Ineq           6.671e+01  2.332e+01   2.860  0.00777 **
## Prob          -4.865e+03  2.316e+03  -2.101  0.04447 *
## Time          -2.638e+00  7.342e+00  -0.359  0.72199
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 211.7 on 29 degrees of freedom
## Multiple R-squared:  0.7089, Adjusted R-squared:  0.5583
## F-statistic: 4.708 on 15 and 29 DF, p-value: 0.000177

confint(lin_mod, level = 0.95)

##              2.5 %              97.5 %
```

```
## (Intercept) -9.109158e+03 -1420.995744
## M          -1.722192e+01  164.486681
## So         -3.016286e+02  315.684357
## Ed          4.008230e+01  304.066872
## Po1        -4.552527e+01  438.267066
## Po2        -3.732969e+02  142.952376
## LF         -3.599503e+03  2504.844767
## M.F        -3.061915e+01   59.591947
## Pop        -4.658764e+00   1.722349
## NW         -9.013851e+00  17.917396
## U1         -1.472868e+04  3139.093150
## U2         -1.201260e+01  338.723164
## Wealth     -1.272025e-01   0.306911
## Ineq        1.901208e+01  114.412062
## Prob       -9.600336e+03 -128.668267
## Time       -1.765331e+01  12.377809
```

If a confidence interval does not include 0, then that means that the predictor IS significant.

Looking at the summary of the linear model of all the predictors, the coefficients that are significant (for $\alpha = 0.05$) are Ed, Ineq, and Prob. The adjusted R-squared is 51.14%, and the model's p-value is 0.001618. Looking at the confidence intervals of the predictors (95% confidence), the significant predictors are Ed, Ineq, and Prob, which are the same compared to the summary. If the confidence interval did not include 0, then that would mean that the predictor is not significant. Therefore, I was able to identify which predictors were significant based on this rule.

```
lin_aic = AIC(lin_mod)
lin_bic = BIC(lin_mod)
lin_aic
```

```
## [1] 623.8799
```

```
lin_bic
```

```
## [1] 654.5932
```

The linear model's BIC is 654.5932.

Corrected AIC formula: $AICc = AIC + 2k(k+1) / (n-k-1)$ where k = number of parameters being estimated and n = number of data points

```
n = length(lin_mod$fitted.values)
k = length(coef(lin_mod))
lin_aicc = lin_aic + (2 * k * (k + 1)) / (n - k - 1)
lin_aicc
```

```
## [1] 643.3085
```

The linear model's Corrected AIC is 596.725.

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
```

```
vif(lin_mod)
```

```
##           M           So           Ed           Po1           Po2           LF           M.F           Pop
## 3.171221 5.242128 4.893921 95.167942 99.532154 3.471818 3.459548 2.764155
##           NW           U1           U2           Wealth           Ineq           Prob           Time
## 4.696039 6.330329 5.257043 9.615647 8.574537 2.721969 2.742236
```

```
library(car)
vif(lin_mod)
```

```
##           M           So           Ed           Po1           Po2           LF           M.F           Pop
## 3.171221 5.242128 4.893921 95.167942 99.532154 3.471818 3.459548 2.764155
##           NW           U1           U2           Wealth           Ineq           Prob           Time
## 4.696039 6.330329 5.257043 9.615647 8.574537 2.721969 2.742236
```

Let's try removing the VIF's that cause multicollinearity. Therefore, I will try removing Po2 since it seems like it's correlated with Po1 and Wealth, which seems like it's correlated with Ineq.

```
test_lin_mod = lm(Crime ~ . - Wealth - Po2, data = crime_data)
summary(test_lin_mod)
```

```
##
## Call:
## lm(formula = Crime ~ . - Wealth - Po2, data = crime_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -458.17 -103.67   4.55  119.36  531.05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.206e+03  1.827e+03  -2.849 0.007719 **
## M             7.091e+01  4.317e+01   1.643 0.110553
## So            4.505e+01  1.455e+02   0.310 0.758866
## Ed            1.697e+02  6.271e+01   2.705 0.010987 *
## Po1           1.009e+02  2.430e+01   4.155 0.000237 ***
## LF           -3.211e+01  1.419e+03  -0.023 0.982097
## M.F           1.572e+01  2.162e+01   0.727 0.472453
## Pop          -1.325e+00  1.541e+00  -0.860 0.396365
## NW            1.621e+00  6.115e+00   0.265 0.792755
```

```
## U1          -5.534e+03  4.246e+03  -1.303  0.202102
## U2           1.721e+02  8.355e+01   2.060  0.047851 *
## Ineq         5.746e+01  1.835e+01   3.131  0.003779 **
## Prob        -4.576e+03  2.236e+03  -2.046  0.049295 *
## Time         9.208e-02  6.917e+00   0.013  0.989465
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 210.1 on 31 degrees of freedom
## Multiple R-squared:  0.6934, Adjusted R-squared:  0.5648
## F-statistic: 5.393 on 13 and 31 DF,  p-value: 5.862e-05
```

```
vif(test_lin_mod)
```

```
##          M          So          Ed          Po1          LF          M.F          Pop          NW
## 3.039109 4.943755 4.690644 4.075756 3.187700 3.373872 2.736700 4.111831
##          U1          U2          Ineq          Prob          Time
## 6.071371 5.066616 5.387688 2.577108 2.470705
```

```
lin_aic1 = AIC(test_lin_mod)
lin_bic1 = BIC(test_lin_mod)
lin_bic1
```

```
## [1] 649.3126
```

```
#Corrected AIC
```

```
n = length(test_lin_mod$fitted.values)
k = length(coef(test_lin_mod))
lin_aicc1 = lin_aic1 + (2 * k * (k + 1)) / (n - k - 1)
lin_aicc1
```

```
## [1] 636.2126
```

After performing VIF on the new model, all of the variables VIF's are close to 5 and lower, which is a good sign. That shows that there is no correlation between the predictor variables, so there is also no interruption with the individual effects to the response variable, Crime.

Based on the p-value tests of each predictor, the significant predictors are Ed, Po1, U2, Ineq, and Prob. Therefore, I will try creating a model with those variables.

```
test_lin_mod1 = lm(Crime ~ Ed + Po1 + U2 + Ineq + Prob, data = crime_data)
summary(test_lin_mod1)
```

```
##
## Call:
## lm(formula = Crime ~ Ed + Po1 + U2 + Ineq + Prob, data = crime_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -458.53 -131.81    1.29  128.41  494.25
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2732.61      864.60  -3.161  0.00304 **
## Ed           140.07       50.04   2.799  0.00792 **
## Po1           96.42       16.98   5.677 1.45e-06 ***
## U2            32.91       41.97   0.784  0.43774
## Ineq          71.96       15.27   4.712 3.09e-05 ***
## Prob        -3926.25     1679.73  -2.337  0.02464 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 217.3 on 39 degrees of freedom
## Multiple R-squared:  0.5872, Adjusted R-squared:  0.5343
## F-statistic: 11.1 on 5 and 39 DF,  p-value: 1.08e-06
```

```
lin_aic2 = AIC(test_lin_mod1)
lin_bic2 = BIC(test_lin_mod1)
lin_bic2
```

```
## [1] 632.237
```

```
#Corrected AIC
```

```
n = length(test_lin_mod1$fitted.values)
k = length(coef(test_lin_mod1))
lin_aicc2 = lin_aic2 + (2 * k * (k + 1)) / (n - k - 1)
lin_aicc2
```

```
## [1] 621.8009
```

As we can see here, the only insignificant predictor is U2. Let's try taking out that variable one last time.

```
test_lin_mod2 = lm(Crime ~ Ed + Po1 + Ineq + Prob, data = crime_data)
summary(test_lin_mod2)
```

```
##
## Call:
## lm(formula = Crime ~ Ed + Po1 + Ineq + Prob, data = crime_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -472.39 -123.79   13.33  131.77  485.07
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2456.77      785.98  -3.126  0.00330 **
## Ed           126.17       46.56   2.710  0.00987 **
## Po1           98.32       16.73   5.878 7.01e-07 ***
## Ineq          70.34       15.06   4.672 3.35e-05 ***
## Prob        -3985.64     1669.92  -2.387  0.02182 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 216.3 on 40 degrees of freedom
## Multiple R-squared:  0.5807, Adjusted R-squared:  0.5388
## F-statistic: 13.85 on 4 and 40 DF,  p-value: 3.554e-07
```

```
vif(test_lin_mod2)
```

```
##          Ed          Po1          Ineq          Prob
## 2.439722 1.823172 3.422788 1.355861
```

```
confint(test_lin_mod2)
```

```
##                2.5 %    97.5 %
## (Intercept) -4045.30161 -868.2455
## Ed           32.06385  220.2711
## Po1           64.51723  132.1323
## Ineq          39.90934  100.7717
## Prob         -7360.68117 -610.6055
```

```
lin_aic3 = AIC(test_lin_mod2)
lin_bic3 = BIC(test_lin_mod2)
lin_bic3
```

```
## [1] 629.1341
```

```
#Corrected AIC
```

```
n = length(test_lin_mod2$fitted.values)
k = length(coef(test_lin_mod2))
lin_aicc3 = lin_aic3 + (2 * k * (k + 1)) / (n - k - 1)
lin_aicc3
```

```
## [1] 619.8326
```

Here is all the information about this model. The P-value is 3.554e-07, which is the lowest that I have received out of all of the linear models I have created. The Adjusted R-squared is 53.88%, which means that 53.88% of the variation is explained by the model with the variables: Ed, Po1, Ineq, and Prob. This current model's Adjusted R-Squared is only slightly less than the other models (maximum 2-3%). The VIF of the predictors are all below 5, which is very good because that means that multicollinearity is not present. All of the predictors are significant because all of their p-values are lower than 0.05. Finally, the Corrected AIC is lower than that of the previous model (619.8326 < 621.8009). The BIC is lower than that of the previous model, but only by ~3 (629.1341 < 632.237). Therefore, this current BIC is 'somewhat likely' to do better than the previous model's. Based on all this information, I believe that this is the best performing model for `lm()`.

Now, we will predict the observed crime rate using the information in the summary of this model.

```
test_point = data.frame(M = 14.0, So = 0, Ed = 10.0, Po1 = 12.0, Po2 = 15.5, LF = 0.640, M.F = 94.0, Po3 = 10.0)

pred_lin_model = predict(test_lin_mod2, test_point)
round(pred_lin_model)
```

```
##      1
## 1239
```

The predicted observed crime rate from the model that consists of Ed, Po1, Ineq, and Prob, is 1239. This means that there were 1239 number of offenses per 100,000 population in 1960.

```
library(DAAG)
```

```
## Warning: package 'DAAG' was built under R version 4.3.3
```

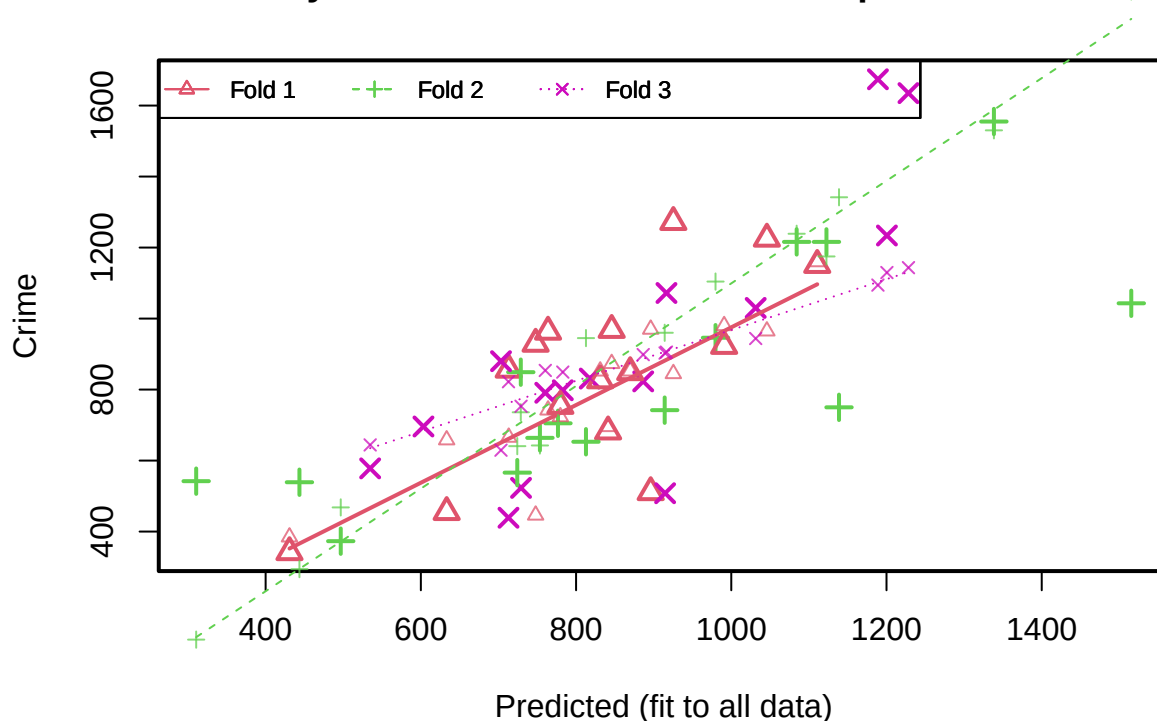
```
##
## Attaching package: 'DAAG'
```

```
## The following object is masked from 'package:car':
##
##      vif
```

```
set.seed(42)
lm_cv = cv.lm(crime_data, test_lin_mod2, n = 4)
```

```
## Warning in cv.lm(crime_data, test_lin_mod2, n = 4):
##
## As there is >1 explanatory variable, cross-validation
## predicted values for a fold are not a linear function
## of corresponding overall predicted values. Lines that
## are shown for the different folds are approximate
```

Small symbols show cross-validation predicted values



```

##
## fold 1
## Observations in test set: 15
##           6           7           9           13           18           20           24
## Predicted  841.27871 763.8739 713.2248 896.1522 747.8857 1045.8536 845.90095
## cvpred     695.63205 742.2205 665.5849 969.8280 446.1906 964.9289 873.22241
## Crime      682.00000 963.0000 856.0000 511.0000 929.0000 1225.0000 968.00000
## CV residual -13.63205 220.7795 190.4151 -458.8280 482.8094 260.0711 94.77759
##           27           32           34           36           39           40
## Predicted  430.74850 779.74655 990.79551 925.3504 831.10889 1110.748513
## cvpred     384.62049 723.25751 980.00619 844.7996 852.03844 1159.975853
## Crime      342.00000 754.00000 923.00000 1272.0000 826.00000 1151.000000
## CV residual -42.62049 30.74249 -57.00619 427.2004 -26.03844 -8.975853
##           45           47
## Predicted  633.2723 869.774413
## cvpred     658.1636 852.169135
## Crime      455.0000 849.000000
## CV residual -203.1636 -3.169135
##
## Sum of squares = 835990.6    Mean square = 55732.71    n = 15
##
## fold 2
## Observations in test set: 15
##           8           10           12           14           16           17
## Predicted  1338.38147 776.6806262 728.7721 753.43337 979.6940 443.5644
## cvpred     1530.22114 705.7713863 736.0036 642.56025 1104.2148 294.2679
## Crime      1555.00000 705.0000000 849.0000 664.00000 946.0000 539.0000
## CV residual 24.77886 -0.7713863 112.9964 21.43975 -158.2148 244.7321
##           19           21           23           28           29           31
## Predicted  1138.802 914.2062 1084.23057 1122.63118 1515.3898 496.78782
## cvpred     1341.677 959.9837 1238.86112 1174.94156 1921.4786 468.08922
## Crime      750.000 742.0000 1216.00000 1216.00000 1043.0000 373.00000
## CV residual -591.677 -217.9837 -22.86112 41.05844 -878.4786 -95.08922
##           35           38           42
## Predicted  812.7626 724.26973 310.52005
## cvpred     945.0022 640.61629 95.64712
## Crime      653.0000 566.00000 542.00000
## CV residual -292.0022 -74.61629 446.35288
##
## Sum of squares = 1569405    Mean square = 104627    n = 15
##
## fold 3
## Observations in test set: 15
##           1           2           3           5           11           15
## Predicted  760.32865 1228.2993 534.68512 1200.5165 1188.9323 782.91679
## cvpred     853.90045 1143.3379 644.07845 1129.6058 1094.3104 849.23208
## Crime      791.00000 1635.0000 578.00000 1234.0000 1674.0000 798.00000
## CV residual -62.90045 491.6621 -66.07845 104.3942 579.6896 -51.23208
##           22           25           30           33           37           41
## Predicted  712.9191 729.0951 603.360301 916.4633 817.673578 703.2720
## cvpred     822.0742 752.9780 692.489126 905.5032 832.208411 629.2631
## Crime      439.0000 523.0000 696.000000 1072.0000 831.000000 880.0000
## CV residual -383.0742 -229.9780 3.510874 166.4968 -1.208411 250.7369
##           43           44           46

```

```
## Predicted    886.5284 1031.45827  914.7109
## cvpred      898.4512  943.86844  902.8479
## Crime       823.0000 1030.00000  508.0000
## CV residual -75.4512   86.13156 -394.8479
##
## Sum of squares = 1058874    Mean square = 70591.57    n = 15
##
## Overall (Sum over all 15 folds)
##      ms
## 76983.77
```

The MSE is 70591.57, which is close to the lowest MSE out of all of the linear models that I have created. Now, let's test out different models using `glm()`.

Poisson model

```
poisson_mod = glm(Crime~., data = crime_data, family = poisson)
summary(poisson_mod)

##
## Call:
## glm(formula = Crime ~ ., family = poisson, data = crime_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -7.496e-01  2.966e-01  -2.527 0.011498 *
## M            8.440e-02  7.343e-03  11.494 < 2e-16 ***
## So           7.862e-02  2.367e-02   3.322 0.000895 ***
## Ed           1.962e-01  1.093e-02  17.958 < 2e-16 ***
## Po1          2.270e-01  1.831e-02  12.393 < 2e-16 ***
## Po2         -1.376e-01  1.972e-02  -6.981 2.94e-12 ***
## LF           1.446e-01  2.343e-01   0.617 0.536969
## M.F          8.875e-03  3.626e-03   2.448 0.014371 *
## Pop         -1.838e-03  2.404e-04  -7.643 2.13e-14 ***
## NW           5.443e-03  1.084e-03   5.023 5.08e-07 ***
## U1          -5.675e+00  6.853e-01  -8.282 < 2e-16 ***
## U2           1.768e-01  1.324e-02  13.354 < 2e-16 ***
## Wealth       1.921e-04  1.657e-05  11.595 < 2e-16 ***
## Ineq         8.964e-02  3.782e-03  23.700 < 2e-16 ***
## Prob        -5.885e+00  3.805e-01 -15.467 < 2e-16 ***
## Time        -2.486e-03  1.203e-03  -2.066 0.038801 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 5051.8  on 44  degrees of freedom
## Residual deviance: 1253.3  on 29  degrees of freedom
## AIC: 1668.9
##
## Number of Fisher Scoring iterations: 4
```

```
confint(poisson_mod)
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %           97.5 %
## (Intercept) -1.330764500 -0.1680707344
## M           0.070006727  0.0987899162
## So          0.032223335  0.1250105813
## Ed          0.174845219  0.2176799505
## Po1         0.191090676  0.2628778684
## Po2        -0.176305380 -0.0990202204
## LF         -0.315152083  0.6031312733
## M.F         0.001765769  0.0159778285
## Pop        -0.002309039 -0.0013665589
## NW          0.003320960  0.0075684569
## U1         -7.018462514 -4.3322891844
## U2          0.150846553  0.2027477389
## Wealth      0.000159648  0.0002245902
## Ineq        0.082224574  0.0970512451
## Prob       -6.632177942 -5.1407074562
## Time       -0.004847459 -0.0001304712
```

```
poisson_aic = AIC(poisson_mod)
poisson_bic = BIC(poisson_mod)
poisson_aic
```

```
## [1] 1668.923
```

```
poisson_bic
```

```
## [1] 1697.83
```

Poisson Model Corrected AIC

```
n = length(poisson_mod$fitted.values)
k = length(coef(poisson_mod))
poisson_aicc = poisson_aic + (2 * k * (k + 1)) / (n - k - 1)
poisson_aicc
```

```
## [1] 1688.352
```

```
#Pseudo R-Squared for Poisson Model
```

```
library(pscl)
```

```
## Warning: package 'pscl' was built under R version 4.3.3
```

```
## Classes and Methods for R originally developed in the
## Political Science Computational Laboratory
## Department of Political Science
## Stanford University (2002-2015),
## by and under the direction of Simon Jackman.
## hurdle and zeroinfl functions by Achim Zeileis.
```

```
pR2(poisson_mod)
```

```
## fitting null model for pseudo-r2
```

```
##          llh          llhNull          G2          McFadden          r2ML
## -818.4617076 -2717.7056852  3798.4879554    0.6988409    1.0000000
##          r2CU
##          1.0000000
```

```
#Poisson cv.glm()
```

```
library(boot)
```

```
## Warning: package 'boot' was built under R version 4.3.3
```

```
##
```

```
## Attaching package: 'boot'
```

```
## The following object is masked from 'package:car':
```

```
##
```

```
##      logit
```

```
poisson_cv = cv.glm(crime_data, poisson_mod, K = 4)
```

```
poisson_cv
```

```
## $call
```

```
## cv.glm(data = crime_data, glmfit = poisson_mod, K = 4)
```

```
##
```

```
## $K
```

```
## [1] 4
```

```
##
```

```
## $delta
```

```
## [1] 112386.99  94722.96
```

```
##
```

```
## $seed
```

```
## [1]          10403          60 1118745441 1531885299    2763106 2140198868
## [7] 1789174487 -845059891 -1092788124 -695153078 -320332875 1768072975
## [13] 1625781270  388718272  642795651 1378608289 -456324464 -949634706
## [19]  376543001  762448811 -707143270 1098218844 1309676639 -302363595
## [25] -1069787188 -1394539742  237698429 1290899063 -814237794 -266332584
## [31] 2139062203 -2033086119  25004776 -705146090 -1756965743 2102593859
## [37] -1873443758 -1694296348 -1047834329 -1442872963 -707344268 2118199898
## [43]  966616741 -1075465025  229050182  559293904 -1608157197 -1169305839
## [49] -16478400 -1680050402 1531439081 -2105537253 -1848449238 -1822459828
```

##	[55]	699682831	-1586685115	-1104835332	1457453906	-1691345715	2114509959
##	[61]	-1991874706	-1192748728	-549462709	1936425129	-1067539976	-379121370
##	[67]	1601917505	367795155	-890494	1259183540	-266262217	-667796435
##	[73]	1164650820	634527338	-1608181547	-411273617	-1526668426	2032800928
##	[79]	1782975843	531406337	-2065834640	-1680261874	-1108410503	-59080949
##	[85]	1143570490	146378556	-1418728321	-1110736939	441889260	2096005954
##	[91]	-75133795	-6687465	1079233662	1425204280	-1374174437	1529358457
##	[97]	1631793352	-675407626	-1873958607	1948066147	252523314	643447940
##	[103]	1416104135	1185166237	-102287596	-2048987718	880757765	-80453153
##	[109]	639378534	-1237687312	-1318249197	546046641	29896032	-1237359874
##	[115]	1414061705	1248070075	-96029814	-1628415380	-1665035281	15588389
##	[121]	-1646361380	1236769458	-165560915	1534099815	-1206149490	1896875880
##	[127]	429670251	-372397815	785301400	-1685993786	69899297	1974780083
##	[133]	-2133550558	-777053164	380506263	-427826547	-1406913628	393989514
##	[139]	1365506549	-553708465	-909814954	-998553088	1006904131	2033230561
##	[145]	-67899312	154882222	-1185135527	-1408912277	1733392858	112411292
##	[151]	808623263	-451637515	-1064324212	1311457890	1629861437	1095227575
##	[157]	-1983158434	134851224	2105063931	578380697	-1558561752	-905597738
##	[163]	2108447057	-88655869	-1304469614	-424227548	1228735463	17012797
##	[169]	1592100532	1499958554	1409572453	490533631	1714481030	1868093712
##	[175]	-1245992525	696644177	507814400	2085961438	946402601	-136185637
##	[181]	-258654230	-1282089076	-1205677873	-1015426939	-2001767620	326404626
##	[187]	-638467827	755075655	1196527662	1713497480	1621458059	1317017193
##	[193]	-1891087816	1225890534	1165948545	1304959507	-1118151230	1659049588
##	[199]	440871031	1812154733	-1107022332	-928031446	-430802283	2122092847
##	[205]	-277846986	-676937632	-744839517	-1231461951	1012915632	1819382990
##	[211]	1974764089	85890123	136771066	-65474308	503129407	351885845
##	[217]	725528876	-1622367742	-410357539	-1502116905	-1025622338	1082575096
##	[223]	-1864272165	-2101785287	-1613211512	1995726902	-922889103	-448979293
##	[229]	-1837902862	-350833732	1025582194	-580171792	841917508	-1822635944
##	[235]	-1142685638	53823600	-1627849156	62547732	116656834	-911846688
##	[241]	1788707964	-1340322608	2039993250	910396968	-118685428	-2059171908
##	[247]	652292402	900323520	1499075124	-1592468792	-2058812742	1477889168
##	[253]	-1124661524	-761375148	234219154	1690281760	-1125713972	-1875056416
##	[259]	-194687294	-1313221592	-626418772	2049395388	64774642	8775216
##	[265]	-485296220	679625144	-816932006	412259472	731028348	-885252332
##	[271]	-1602560702	-1045885184	118307772	1287855952	-1812733662	1708829256
##	[277]	-711649268	-1731425188	-878341678	268932608	-46386860	-1980555032
##	[283]	-793423622	1457612496	182995244	1399060724	542240146	363956576
##	[289]	-1664546804	1806183136	1439310434	505985704	-204326324	-943601796
##	[295]	1486572850	136324720	680311620	226478040	-2112856198	-746033616
##	[301]	384210876	214275668	-738244542	157696160	168570556	-739948080
##	[307]	103031650	2085995240	-672972596	1679671804	-495162894	-333104512
##	[313]	2006021620	288597640	-1474060486	-1131855024	1825322092	80034196
##	[319]	-363404206	-1834145312	1059781836	425405600	2144654722	-1925996312
##	[325]	-597623060	805473596	1638825074	-1888886672	1713535012	-854873032
##	[331]	524384026	1165376720	-1603774660	-1742350444	-243643518	-1676725824
##	[337]	-599896516	1629332880	1433134626	-853533432	-1707571572	-2145528036
##	[343]	631546002	206145920	-887300588	1347489576	1910297146	1959318992
##	[349]	1974569324	479667124	1560446994	1548922336	1439170508	-240983840
##	[355]	-228632542	-1949089496	-604291572	-993566916	712752114	611658480
##	[361]	-1530124732	-1961008552	-214278342	605992432	1610452668	-1605838572
##	[367]	-571262270	1835437152	-1029983364	-1924954160	647430434	1655261864
##	[373]	1063711628	1295165628	-2048596174	-1336047040	850044084	-2035320248

```
## [379] -1595384390 754465296 1210551020 -19609260 336183570 1891581344
## [385] 1308957388 -922648992 1012277186 731954984 -433008724 -1528614084
## [391] -2132768398 1989884720 -132594396 -1074668744 546068186 -75735152
## [397] 363376380 -409697388 1542830146 -847174400 351290428 530448720
## [403] 1947601058 -1451402936 1974634252 -844025124 -904532654 1049345152
## [409] -494288812 803346408 -1075715846 561220432 2031509292 198154612
## [415] 7261074 -736253856 440640908 -1369654432 673090018 216673192
## [421] 1370209484 -1151535108 1948247474 767626736 740801476 -45179560
## [427] 1443639674 413125552 -250478148 1236666452 2112745410 194607136
## [433] -1478371140 345768528 -1428059166 -7526296 1998830668 -1712899844
## [439] 1721400306 -1334372096 -428162700 -1973943288 2008448954 -1026813488
## [445] -1535599636 926136724 -1718222382 1189269856 1782738764 1784391968
## [451] -1777002622 797691624 1396777196 918512700 -1646167182 1409369712
## [457] -1526359844 -842789043 -579076425 -885658560 -14670562 1202733035
## [463] 776307565 791695834 1091701656 -988630527 -1501289549 -1673674028
## [469] -2058050590 -299862409 1851551681 806096198 -1240032124 -148299211
## [475] 1539598879 -224514696 -1044894634 1977888995 1537611429 1813502786
## [481] -125718224 -1790113447 177933003 1542550332 -1729209494 1510750623
## [487] -431411159 148540094 1629670316 -1227687683 2052748359 737481520
## [493] 1891341390 2053503099 652327901 -127213046 -1920694488 549099921
## [499] -1879728253 -1644374908 884675250 1413662215 2055882321 1467940310
## [505] -1381150540 1188072645 674778639 85149320 -787862618 2045961043
## [511] 1661480117 -1659090414 440141792 -641852855 190976123 1116946764
## [517] 638997722 -1595108145 1669972121 762240174 -1379505732 -1152036307
## [523] 1609049559 -2038076384 190828926 73826123 -450050739 -2048816262
## [529] 1252396472 40709985 294979347 -1910300876 -932262590 -165019305
## [535] 1056033953 1317655910 1273608036 1007974037 -315314241 -92376232
## [541] 1769460086 1908593539 -450915899 -1932937502 162912080 1561162361
## [547] -366463573 -2022687972 260017226 -1632184257 -615937911 1495796894
## [553] 1434925388 -1307713251 -444601177 -536020016 -1739586706 -136248805
## [559] 1499923325 551035242 -1716477944 992823473 -1883396317 1807328292
## [565] 171112146 -1023323353 -1668983439 229776310 -396017836 -2008434203
## [571] 1667720175 117191464 954872710 -194637901 51226389 -838020750
## [577] 766088640 -1863989719 1868012187 -1699290260 -401343622 126092463
## [583] -556421191 499706318 -1456681700 -1118001011 1331725559 905781248
## [589] -268058018 418159915 1591369261 -42129766 -831327656 552078785
## [595] 67170931 1109389204 -951255518 -300400329 1128910849 -986632954
## [601] 392796484 1919208565 -1094820385 -39343816 -1406786794 453294883
## [607] 1937314021 -440537854 -1688101008 -1330069735 1554044683 -936823812
## [613] -1358000086 -1652868001 443773929 200916222 -1119456020 1815619645
## [619] 210361351 1057458160 -1116523634 -388009541 -1615335523 -205931190
## [625] -542558872 -1549119187
```

```
mse_raw = poisson_cv$delta[1]
mse_adjusted = poisson_cv$delta[2]
c(mse_raw, mse_adjusted)
```

```
## [1] 112386.99 94722.96
```

```
vif(poisson_mod)
```

```
##      M      So      Ed      Po1      Po2      LF      M.F      Pop      NW      U1
## 3.5098 4.9514 5.1413 92.2440 98.7120 3.0568 3.5288 2.9019 4.4017 5.5432
```

```
##      U2  Wealth    Ineq    Prob    Time
##  4.8185  8.3613  8.1423  2.5683  2.7430
```

```
poisson_mod1 = glm(Crime ~ . - Po2 - Wealth - NW - Time, data = crime_data, family = poisson)
summary(poisson_mod1)
```

```
##
## Call:
## glm(formula = Crime ~ . - Po2 - Wealth - NW - Time, family = poisson,
##      data = crime_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.3500465  0.2753877  -1.271 0.203692
## M            0.0818361  0.0064355  12.716 < 2e-16 ***
## So           0.1296196  0.0213963   6.058 1.38e-09 ***
## Ed           0.1870773  0.0105202  17.783 < 2e-16 ***
## Po1          0.1196755  0.0036174  33.083 < 2e-16 ***
## LF           0.8117080  0.2177230   3.728 0.000193 ***
## M.F          0.0138258  0.0033709   4.102 4.10e-05 ***
## Pop         -0.0014933  0.0002226  -6.710 1.95e-11 ***
## U1          -5.4376230  0.6597405  -8.242 < 2e-16 ***
## U2           0.1909822  0.0129679  14.727 < 2e-16 ***
## Ineq         0.0674411  0.0029484  22.874 < 2e-16 ***
## Prob        -5.6884552  0.3194308 -17.808 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 5051.8  on 44  degrees of freedom
## Residual deviance: 1437.4  on 33  degrees of freedom
## AIC: 1845
##
## Number of Fisher Scoring iterations: 4
```

```
poisson_cv1 = cv.glm(crime_data, poisson_mod1, K = 4)
poisson_cv1
```

```
## $call
## cv.glm(data = crime_data, glmfit = poisson_mod1, K = 4)
##
## $K
## [1] 4
##
## $delta
## [1] 71905.26 63911.69
##
## $seed
## [1]      10403      129 1118745441 1531885299      2763106 2140198868
## [7] 1789174487 -845059891 -1092788124 -695153078 -320332875 1768072975
## [13] 1625781270 388718272 642795651 1378608289 -456324464 -949634706
## [19] 376543001 762448811 -707143270 1098218844 1309676639 -302363595
```

##	[25]	-1069787188	-1394539742	237698429	1290899063	-814237794	-266332584
##	[31]	2139062203	-2033086119	25004776	-705146090	-1756965743	2102593859
##	[37]	-1873443758	-1694296348	-1047834329	-1442872963	-707344268	2118199898
##	[43]	966616741	-1075465025	229050182	559293904	-1608157197	-1169305839
##	[49]	-16478400	-1680050402	1531439081	-2105537253	-1848449238	-1822459828
##	[55]	699682831	-1586685115	-1104835332	1457453906	-1691345715	2114509959
##	[61]	-1991874706	-1192748728	-549462709	1936425129	-1067539976	-379121370
##	[67]	1601917505	367795155	-890494	1259183540	-266262217	-667796435
##	[73]	1164650820	634527338	-1608181547	-411273617	-1526668426	2032800928
##	[79]	1782975843	531406337	-2065834640	-1680261874	-1108410503	-59080949
##	[85]	1143570490	146378556	-1418728321	-1110736939	441889260	2096005954
##	[91]	-75133795	-6687465	1079233662	1425204280	-1374174437	1529358457
##	[97]	1631793352	-675407626	-1873958607	1948066147	252523314	643447940
##	[103]	1416104135	1185166237	-102287596	-2048987718	880757765	-80453153
##	[109]	639378534	-1237687312	-1318249197	546046641	29896032	-1237359874
##	[115]	1414061705	1248070075	-96029814	-1628415380	-1665035281	15588389
##	[121]	-1646361380	1236769458	-165560915	1534099815	-1206149490	1896875880
##	[127]	429670251	-372397815	785301400	-1685993786	69899297	1974780083
##	[133]	-2133550558	-777053164	380506263	-427826547	-1406913628	393989514
##	[139]	1365506549	-553708465	-909814954	-998553088	1006904131	2033230561
##	[145]	-67899312	154882222	-1185135527	-1408912277	1733392858	112411292
##	[151]	808623263	-451637515	-1064324212	1311457890	1629861437	1095227575
##	[157]	-1983158434	134851224	2105063931	578380697	-1558561752	-905597738
##	[163]	2108447057	-88655869	-1304469614	-424227548	1228735463	17012797
##	[169]	1592100532	1499958554	1409572453	490533631	1714481030	1868093712
##	[175]	-1245992525	696644177	507814400	2085961438	946402601	-136185637
##	[181]	-258654230	-1282089076	-1205677873	-1015426939	-2001767620	326404626
##	[187]	-638467827	755075655	1196527662	1713497480	1621458059	1317017193
##	[193]	-1891087816	1225890534	1165948545	1304959507	-1118151230	1659049588
##	[199]	440871031	1812154733	-1107022332	-928031446	-430802283	2122092847
##	[205]	-277846986	-676937632	-744839517	-1231461951	1012915632	1819382990
##	[211]	1974764089	85890123	136771066	-65474308	503129407	351885845
##	[217]	725528876	-1622367742	-410357539	-1502116905	-1025622338	1082575096
##	[223]	-1864272165	-2101785287	-1613211512	1995726902	-922889103	-448979293
##	[229]	-1837902862	-350833732	1025582194	-580171792	841917508	-1822635944
##	[235]	-1142685638	53823600	-1627849156	62547732	116656834	-911846688
##	[241]	1788707964	-1340322608	2039993250	910396968	-118685428	-2059171908
##	[247]	652292402	900323520	1499075124	-1592468792	-2058812742	1477889168
##	[253]	-1124661524	-761375148	234219154	1690281760	-1125713972	-1875056416
##	[259]	-194687294	-1313221592	-626418772	2049395388	64774642	8775216
##	[265]	-485296220	679625144	-816932006	412259472	731028348	-885252332
##	[271]	-1602560702	-1045885184	118307772	1287855952	-1812733662	1708829256
##	[277]	-711649268	-1731425188	-878341678	268932608	-46386860	-1980555032
##	[283]	-793423622	1457612496	182995244	1399060724	542240146	363956576
##	[289]	-1664546804	1806183136	1439310434	505985704	-204326324	-943601796
##	[295]	1486572850	136324720	680311620	226478040	-2112856198	-746033616
##	[301]	384210876	214275668	-738244542	157696160	168570556	-739948080
##	[307]	103031650	2085995240	-672972596	1679671804	-495162894	-333104512
##	[313]	2006021620	288597640	-1474060486	-1131855024	1825322092	80034196
##	[319]	-363404206	-1834145312	1059781836	425405600	2144654722	-1925996312
##	[325]	-597623060	805473596	1638825074	-1888886672	1713535012	-854873032
##	[331]	524384026	1165376720	-1603774660	-1742350444	-243643518	-1676725824
##	[337]	-599896516	1629332880	1433134626	-853533432	-1707571572	-2145528036
##	[343]	631546002	206145920	-887300588	1347489576	1910297146	1959318992

```
## [349] 1974569324 479667124 1560446994 1548922336 1439170508 -240983840
## [355] -228632542 -1949089496 -604291572 -993566916 712752114 611658480
## [361] -1530124732 -1961008552 -214278342 605992432 1610452668 -1605838572
## [367] -571262270 1835437152 -1029983364 -1924954160 647430434 1655261864
## [373] 1063711628 1295165628 -2048596174 -1336047040 850044084 -2035320248
## [379] -1595384390 754465296 1210551020 -19609260 336183570 1891581344
## [385] 1308957388 -922648992 1012277186 731954984 -433008724 -1528614084
## [391] -2132768398 1989884720 -132594396 -1074668744 546068186 -75735152
## [397] 363376380 -409697388 1542830146 -847174400 351290428 530448720
## [403] 1947601058 -1451402936 1974634252 -844025124 -904532654 1049345152
## [409] -494288812 803346408 -1075715846 561220432 2031509292 198154612
## [415] 7261074 -736253856 440640908 -1369654432 673090018 216673192
## [421] 1370209484 -1151535108 1948247474 767626736 740801476 -45179560
## [427] 1443639674 413125552 -250478148 1236666452 2112745410 194607136
## [433] -1478371140 345768528 -1428059166 -7526296 1998830668 -1712899844
## [439] 1721400306 -1334372096 -428162700 -1973943288 2008448954 -1026813488
## [445] -1535599636 926136724 -1718222382 1189269856 1782738764 1784391968
## [451] -1777002622 797691624 1396777196 918512700 -1646167182 1409369712
## [457] -1526359844 -842789043 -579076425 -885658560 -14670562 1202733035
## [463] 776307565 791695834 1091701656 -988630527 -1501289549 -1673674028
## [469] -2058050590 -299862409 1851551681 806096198 -1240032124 -148299211
## [475] 1539598879 -224514696 -1044894634 1977888995 1537611429 1813502786
## [481] -125718224 -1790113447 177933003 1542550332 -1729209494 1510750623
## [487] -431411159 148540094 1629670316 -1227687683 2052748359 737481520
## [493] 1891341390 2053503099 652327901 -127213046 -1920694488 549099921
## [499] -1879728253 -1644374908 884675250 1413662215 2055882321 1467940310
## [505] -1381150540 1188072645 674778639 85149320 -787862618 2045961043
## [511] 1661480117 -1659090414 440141792 -641852855 190976123 1116946764
## [517] 638997722 -1595108145 1669972121 762240174 -1379505732 -1152036307
## [523] 1609049559 -2038076384 190828926 73826123 -450050739 -2048816262
## [529] 1252396472 40709985 294979347 -1910300876 -932262590 -165019305
## [535] 1056033953 1317655910 1273608036 1007974037 -315314241 -92376232
## [541] 1769460086 1908593539 -450915899 -1932937502 162912080 1561162361
## [547] -366463573 -2022687972 260017226 -1632184257 -615937911 1495796894
## [553] 1434925388 -1307713251 -444601177 -536020016 -1739586706 -136248805
## [559] 1499923325 551035242 -1716477944 992823473 -1883396317 1807328292
## [565] 171112146 -1023323353 -1668983439 229776310 -396017836 -2008434203
## [571] 1667720175 117191464 954872710 -194637901 51226389 -838020750
## [577] 766088640 -1863989719 1868012187 -1699290260 -401343622 126092463
## [583] -556421191 499706318 -1456681700 -1118001011 1331725559 905781248
## [589] -268058018 418159915 1591369261 -42129766 -831327656 552078785
## [595] 67170931 1109389204 -951255518 -300400329 1128910849 -986632954
## [601] 392796484 1919208565 -1094820385 -39343816 -1406786794 453294883
## [607] 1937314021 -440537854 -1688101008 -1330069735 1554044683 -936823812
## [613] -1358000086 -1652868001 443773929 200916222 -1119456020 1815619645
## [619] 210361351 1057458160 -1116523634 -388009541 -1615335523 -205931190
## [625] -542558872 -1549119187
```

```
mse_raw = poisson_cv1$delta[1]
mse_adjusted = poisson_cv1$delta[2]
c(mse_raw, mse_adjusted)
```

```
## [1] 71905.26 63911.69
```

```
poisson_aic1 = AIC(poisson_mod1)
poisson_bic1 = BIC(poisson_mod1)
poisson_aic1
```

```
## [1] 1845.025
```

```
poisson_bic1
```

```
## [1] 1866.705
```

```
n = length(poisson_mod1$fitted.values)
k = length(coef(poisson_mod1))
poisson_aicc1 = poisson_aic1 + (2 * k * (k + 1)) / (n - k - 1)
poisson_aicc1
```

```
## [1] 1854.775
```

```
pR2(poisson_mod1)
```

```
## fitting null model for pseudo-r2
```

```
##          llh          llhNull          G2          McFadden          r2ML
## -910.5122755 -2717.7056852  3614.3868194    0.6649702    1.0000000
##          r2CU
##    1.0000000
```

Comparing the two poisson models that I have created, the original poisson model has better traits. The first model has a lower BIC, lower corrected AIC, a slightly larger R-Squared (from the McFadden), and a lower MSE. The only thing is the first model includes all of the predictors, and some predictors are insignificant while the other model's predictors are all significant. If we're looking for predictive accuracy, we would use the first model. Therefore, I will predict the values according to the first model.

```
test_point = data.frame(M = 14.0, So = 0, Ed = 10.0, Po1 = 12.0, Po2 = 15.5, LF = 0.640, M.F = 94.0, Po)
```

```
pred_poisson_model = predict(poisson_mod, test_point, type = 'response')
pred_poisson_model
```

```
##          1
## 293.3126
```

This prediction value looks highly unreasonable, considering the fact that the poisson model is performed on a different scale. Therefore, the multiple linear regression model is a better model to use.